FairChild Institute

AI Software Engineering Institute

12-Month Practical Software Engineering Mentorship Programme

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Phase 1

AI-Powered Full-Stack Development

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Month 1

Computing, Web & Digital Creation

Phase: CREATOR | Theme: From User to Creator

Duration 4 weeks

Month Journey

Week	Theme	Core Question	Identity Shift
1	Meet the Machine	How does a computer actually work?	I can understand the machine I use.
2	Enter the Web	How does the web actually work?	Websites are software, not magic.
3	Build Something Real	How does the web actually work?	I can build something real.
4	Become the Builder	Can I build something without being told every step?	I can figure out how to build.

Final Project Youth Organisation Website

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Month 2

CSS, HTML, Design & Responsive Interfaces

Phase: CREATOR | Theme: From Structure to Experience

Duration 4 weeks

Month Journey

Week	Theme	Build	Struggle	Independence
1	Make It Look Right	Styled interface	Spacing & styling problems	Follow → modify
2	Make It Fit Together	Responsive layouts	Layout decisions	Modify → solve
3	Make It Work Everywhere	Responsive interface	Different screens/ users	Solve with guidance
4	Make It Better	Improved real-world interface	Open-ended design problem	Supported → independent

Final Project Youth Organisation Website

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Month 3

JavaScript & Interactive Applications

Phase: CREATOR | Theme: From Interface to Behaviour

Duration 4 weeks

Month Journey

Week	Theme	Build	Struggle	Independence
1	Think in JavaScript	Small browser programs	Logic & syntax errors	Follow → modify
2	Make the Page Respond	Interactive webpage	DOM & events	Modify → solve
3	Build a Small Application	Form/data-driven app	State, validation & edge cases	Solve with guidance
4	Become the Developer	Open-ended interactive app	Unfamiliar behaviour	Supported → independent

Final Project Interactive Web Application

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Month 4

Programming Foundations & Problem Solving

Phase: PROGRAMMER | Theme: From Writing Code to Solving Problems

Duration 4 weeks

Month Journey

Week	Theme	Build	Struggle	Independence
1	Think Before You Code	Algorithmic solutions	Turning requirements into steps	Follow → modify
2	Build with Structure	Reusable programs	Functions, data & abstraction	Modify → solve
3	Solve & Improve	Problem-solving systems	Edge cases & trade-offs	Solve with guidance
4	Think Like a Programmer	Open-ended interactive app	Unfamiliar requirements	Supported → independent

Final Project Programming Problem Solver

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Month 5

Python, Automation & Programming Transfer

Phase: PROGRAMMER | Theme: From Writing Code to Solving Problems

Duration 4 weeks

Month Journey

Week	Theme	Build	Struggle	Independence
1	Meet Python	Small Python programs	New syntax & errors	Transfer → modify
2	Work With Real Data	Data/file processors	Data representation & invalid data	Build with guidance
3	Automation & Useful Programs	Automation scripts	Safety, side effects & failures	Solve with reduced support
4	Programming Transfer Challenge	Open-ended Python tool	Unfamiliar behaviour	Supported → independent

Final Project Python Automation & Data Tool

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Month 6

Algorithms, Data Structures & Computational Thinking

Phase: PROGRAMMER | Theme: From Writing Solutions to Reasoning About Solutions

Duration 4 weeks

Month Journey

Week	Theme	Primary Capability	Independence
1	Think in Algorithms	Turn problems into procedures	Level 2
2	Choose the Right Data Structure	Represent information to support operations	Level 2 → 3
3	Search, Sort & Compare	Reason about efficiency and trade-offs	Level 3
4	Solve & Defend	Choose, implement and justify an approach	Level 3

Final Project Problem-Solving Algorithm Toolkit

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Month 7

Backend, HTTP & APIs

Phase: SOFTWARE BUILDER | Theme: From Building Interfaces to Building Services

Duration	4 weeks

Month Journey

Week	Theme	Primary Capability
1	Understand the Backend	I can explain how a client and server communicate.
2	Build HTTP APIs	I can create and test API endpoints.
3	Make APIs Reliable	I can validate input, handle errors and debug backend behaviour.
4	Build a Real Service	I can design and build a small backend for a real problem.

Final Project	Backend Service for a Real Problem

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Month 8

PostgreSQL, SQL & DATA Modelling

Phase: SOFTWARE BUILDER | Theme: From Building Services to Managing Data

Duration	4 weeks

Month Journey

Week	Theme	Primary Capability
1	Understand Data	I can explain relational databases and model simple data.
2	Speak SQL	I can create, query and modify relational data with SQL.
3	Connect the Application	I can connect my backend API to PostgreSQL safely.
4	Design Reliable Data	I can design a database for a real application and defend my decisions.

Final Project	Database-Backed Application Service

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Month 9

Full-Stack Product Engineering

Phase: SOFTWARE BUILDER | Theme: From Building Components to Building Products

Duration 4 weeks

Month Journey

Week	Theme	Primary Capability
1	Think in Products	I can turn a real problem into a small full-stack product plan.
2	Build the Frontend	I can build a React + TypeScript interface that consumes an API.
3	Connect the System	I can integrate frontend, backend and PostgreSQL into one application.
4	Ship the Product	I can test, refine, explain and present a complete application.

Final Project Full-Stack Product

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Month 10

Professional Software Engineering

Phase: ENGINEER | Theme: From Building Products to Engineering Software

Duration 4 weeks

Month Journey

Week	Theme	Primary Capability
1	Work With Change	I can use Git and GitHub to manage software changes.
2	Test & Debug	I can use tests and investigation to improve software safely.
3	Review & Refactor	I can review code, improve its structure and work with feedback.
4	Deploy & Maintain	I can prepare, deploy and maintain a software project professionally.

Final Project Professionalise a Full-Stack Product

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Month 11

Security, System & AI-Native Engineering

Phase: ENGINEER | Theme: From Engineering Applications to Understanding the Environment They Run In

Duration 4 weeks

Month Journey

Week	Theme	Primary Capability
1	Think in Products	I can recognise and reduce common application security risks.
2	Build the Frontend	I can explain how software uses operating-system and network resources.
3	Connect the System	I can reason about basic reliability/scalability issues and use AI with verification.
4	Ship the Product	I can investigate a realistic system, identify risks and propose evidence-based improvements.

Final Project Secure, Understand & Improve

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Month 12

Independent Engineering & Capstone

Phase: ENGINEER | Theme: From Guided Building to Independent Engineering

Duration 4 weeks

Month Journey

Week	Theme	Primary Capability
1	Discover & Define	I can identify a problem, research it and define a realistic MVP.
2	Design & Plan	I can design a solution and determine what I need to learn.
3	Build & Investigate	I can implement, research and debug with decreasing mentor support.
4	Refine, Deploy & Defend	I can test, improve, deploy and explain the solution independently.

Final Project Independent Engineering Capstone

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Phase 2

AI-Augmented Software Engineering & Product Development

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Month 1

Reading Real Codebases

Phase: ENGINEER | Theme: From Building Applications to Understanding Existing Software

Duration	4 weeks

Month Journey

- **Week 1: Orient & Run:** Learn the repository, follow setup instructions, run the project and record its moving parts.
- **Week 2: Map the System:** Identify boundaries, dependencies, configuration and the responsibilities of major folders/modules.
- **Week 3: Trace Real Behaviour:** Follow one feature end to end, inspect data flow and investigate one behaviour or bug.
- **Week 4: Make a Safe Change:** Implement a small scoped improvement, verify it, review it and defend the investigation.

Final Project	Codebase Investigation Dossier & Reviewed Change

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Month 2

Advanced TypeScript and Maintainable Code

Phase: ENGINEER | Theme: From Understanding Existing Code to Improving Its Design

Duration 4 weeks

Month Journey

- **Week 1:** Type Models and Function Contracts. Make data shapes and behaviour expectations explicit.
- **Week 2:** Narrowing, Generics and Safe Boundaries. Represent uncertainty deliberately and validate external data.
- **Week 3:** Modules, Responsibilities and Refactoring. Improve code organisation without a rewrite.
- **Week 4:** Verification, Review and Maintainability. Deliver and defend a focused refactor.

Final Project Typed and Maintainable Feature

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Month 3

Frontend Architecture and Product Experience

Phase: ENGINEER | Theme: From Maintainable Code to Coherent Product Interfaces

Duration 4 weeks

Month Journey

- **Week 1:** Component Architecture and State Ownership. Define responsibilities and decide where state belongs.
- **Week 2:** Async Data and User Feedback. Model loading, success, empty, error and retry states.
- **Week 3:** Forms, Validation and Accessibility. Build clear, accessible interactions and predictable validation.
- **Week 4:** Integration, Testing and Product Polish. Verify flows, handle edge cases and defend architectural choices.

Final Project Resilient Frontend Feature

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Month 4

Backend Architecture and API Design

Phase: ENGINEER | AI-Augmented Software Engineering & Product Development

Programme principle: Fundamentals First. AI Amplified. Human Led.

Duration	4 weeks

Month Journey

Week	Theme	Evidence of learning
1	API Contracts and Request Lifecycle	Endpoint map, request-flow sketch, contract decisions
2	Application Boundaries and Business Logic	Refactored service with clear responsibilities
3	Validation, Errors, Security and Access	Consistent error contract and protected routes
4	API Quality, Documentation and Design Review	Tested and documented backend mini-product

Final Project	Well-Designed Resource API

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Month 5

PostgreSQL Data Engineering

Programme principle: Fundamentals First. AI Amplified. Human Led.

Duration	4 weeks

Month Journey

Week	Theme	Evidence of learning
1	Relational Modelling and Data Integrity	ER diagram, schema, constraints and seed data
2	SQL for Real Product Questions	Query collection with joins, aggregation and pagination
3	Transactions, Migrations and Application Data Access	Migration history, transaction exercise and safe application queries
4	Query Plans, Indexes and Data Quality	Measured query investigation and final database-backed service

Final Project	Data-Driven Product API

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Month 6

Testing and Quality Engineering

Programme principle: Fundamentals First. AI Amplified. Human Led.

Duration 4 weeks

Month Journey

Week	Theme	Evidence of learning
1	Test Design and Unit Testing	Risk-based test plan and focused unit tests
2	Integration Testing for APIs and Data	API and database integration test suite
3	End-to-End, Security and Failure Testing	Critical journey tests and failure-path evidence
4	Test Reliability, Coverage and CI	Stable test suite, CI workflow, and quality review

Final Project Regression-Resistant Product

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Month 7

Application Security And Identity

Duration 4 weeks

Month Journey

Week	Theme	Build focus	Evidence
1	Security thinking and identity fundamentals	Threat model, identity flow, access matrix	Threat model and testable rules
2	Authentication and credential handling	Registration/login, password hashing, session or token flow	Authentication tests and flow diagram
3	Authorization and resource ownership	Roles, permissions, ownership checks, safe errors	Protected API and denial-path tests
4	Security verification and hardening	Stable test suite, CI workflow, and quality review	Security evidence pack and demo

Final Project Secure Resource API

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Month 8

Git Collaboration And Engineering Workflow

Duration 4 weeks

Month Journey

Week	Theme	Build focus	Evidence
1	Git model and safe branch workflow	Branches, commits, remotes, synchronization	Clean branch and meaningful commit history
2	Pull requests and code review	PR description, review, feedback, traceability	Reviewed feature pull request
3	Merge conflicts and history safety	Conflict resolution, merge/ rebase, revert/reset	Conflict resolution record and recovery exercise
4	Team workflow simulation	Issues, CI checks, review etiquette, release-ready changes	Team delivery simulation and retrospective

Final Project Reviewable Team Feature

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Month 9

Deployment, CI/CD And Cloud Operations

Duration 4 weeks

Month Journey

Week	Theme	Primary evidence
1	Deployment foundations and environments	Environment map, deployment diagram, first deployed service.
2	Build pipelines and continuous integration	CI workflow that runs automated checks on changes.
3	Configuration, secrets and release safety	Environment configuration, secret-handling review, release checklist.
4	Cloud operations, troubleshooting and delivery project	Deployed application, smoke-test record, operational runbook and demo.

Final Project Repeatable Delivery Pipeline

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Month 10

Performance, Reliability And Observability

Duration 4 weeks

Month Journey

Week	Theme	Primary evidence
1	Performance fundamentals and measurement	Baseline report and reproducible test method.
2	Observability: logs, metrics and health	Useful logs, health check, and a small dashboard or metrics summary.
3	Reliability patterns and failure handling	Timeout/retry decisions, graceful failure, and failure-mode tests.
4	Performance investigation and improvement project	Measured improvement report, operational notes, and demo.

Final Project Measured Performance and Reliability Improvement

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Month 11

AI-Augmented Product Engineering

Duration	4 weeks

Month Journey

Week	Theme	Primary evidence
1	Requirements, Planning and AI Collaboration	Turn a product request into testable requirements and a human-owned implementation plan.
2	AI-Assisted Implementation and Debugging	Implement a bounded feature with AI assistance while preserving code comprehension and evidence-led debugging.
3	Testing, Review and Responsible AI Use	Verify AI-assisted work with tests, code review, security checks, and transparent records.
4	AI-Augmented Product Delivery	Deliver a coherent feature through planning, implementation, testing, review, and documentation, with independent ownership.

Final Project	AI-Assisted Feature Delivery

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Month 12

Product Ownership Capstone

Duration 4 weeks

Month Journey

Week	Theme	Primary evidence
1	Product Definition and Architecture	The learner defines a bounded product problem, establishes acceptance criteria, and proposes an architecture that fits the scope.
2	Build the Core Product	Implement a coherent end-to-end journey with maintainable code and data integrity.
3	Quality, Security and Release Readiness	Verify AI-assisted work with tests, code review, security checks, and transparent records.
4	Demo, Viva and Engineering Reflection	Demonstrate the product, defend engineering decisions, and reflect on readiness for Year 3.

Final Project Phase 2 Product Ownership Capstone

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Phase 3

AI-Driven Systems Architecture & Technical Leadership

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Month 1

Systems Thinking and Architecture Foundations

Duration 4 weeks

Month Journey

Week	Focus	Evidence
1	Systems thinking and boundaries	Context diagram, requirements and assumptions register.
2	Quality attributes and tradeoffs	Scenarios, tradeoff matrix and first ADR.
3	Architecture views and data flow	Container/component views, data flow and traceability.
4	Architecture brief and design review	Reviewed brief, risk register, validation plan and viva.

Final Project Architecture Brief for a Real Product

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Month 2

Architecture Patterns and Modular System Design

Duration 4 weeks

Month Journey

- **Week 1:** Modularity, Coupling and Cohesion. **Deliverable:** module map, contracts and change-impact notes.
- **Week 2:** Architectural Patterns and Their Tradeoffs. **Deliverable:** pattern comparison matrix and architecture decision record..
- **Week 3:** Dependency Direction, Data Ownership and Integration. **Deliverable:** dependency graph, data ownership table, integration contract and fitness check.
- **Week 4:** Apply, Validate and Communicate the Design. **Deliverable:** final modular architecture pack and design review.

Final Project Modular Architecture Pack

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Month 3

Distributed Systems and Service Communication

Duration 4 weeks

Month Journey

- **Week 1:** Distributed-System Fundamentals and Network Reality. **Deliverable:** communication comparison and request/event flow.
- **Week 2:** Delivery Semantics, Retries and Idempotency. **Deliverable:** idempotency contract, retry policy and recovery flow.
- **Week 3:** Consistency, Coordination and Resilience. **Deliverable:** consistency requirements, compensation paths and observability map.
- **Week 4:** Design and Validate a Distributed Workflow. **Deliverable:** final distributed design pack and prototype or simulation.

Final Project Resilient Distributed Workflow

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Month 4

Data-Intensive Systems and Data Architecture

Duration 4 weeks

Month Journey

- **Week 1:** Data Architecture and Storage Tradeoffs. **Deliverable:** Data requirements, ownership map and access-pattern inventory.
- **Week 2:** Query Performance, Caching and Materialised Views. **Deliverable:** Query evidence, caching policy and read-model strategy.
- **Week 3:** Pipelines, Events and Data Consistency. **Deliverable:** Ppeline, data-quality rules, reconciliation plan and recovery checklist.
- **Week 4:** Architecture Decisions and Validation. **Deliverable:** Final data architecture pack, decision record and test plan.

Final Project Data Architecture and Access Strategy

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Month 5

Cloud Architecture and Infrastructure Design

Duration	4 weeks

Month Journey

- **Week 1:** Cloud Foundations and Infrastructure Components. **Deliverable:** Component map, request path and responsibility matrix.
- **Week 2:** Cloud Networking, Identity and Environment Design. **Deliverable:** Network boundaries, role matrix and environment promotion plan.
- **Week 3:** Infrastructure as Code, Deployment and Cost. **Deliverable:** Infrastructure example, deployment workflow and cost assumptions.
- **Week 4:** Reliability, Recovery and Architecture Review. **Deliverable:** Final cloud architecture pack and review.

Final Project	Cloud Deployment Architecture Pack

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Month 6

Reliability Engineering, Observability and Resilience

Duration 4 weeks

Month Journey

- **Week 1:** Reliability Goals and Service Health. **Deliverable**: User-journey map, SLI/SLO table and dependency map.
- **Week 2:** Observability and Diagnosis. **Deliverable**: Telemetry plan, dashboard sketch, alert catalogue and incident diagnosis.
- **Week 3:** Resilience Patterns and Failure Handling. **Deliverable:** Timeout/retry policy, resilience pattern choices and overload plan.
- **Week 4:** Incident Response, Recovery and Reliability Review. **Deliverable:** Runbook, post-incident review and final reliability pack.

Final Project Reliability and Observability Pack

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Month 7

Application Security Architecture and Threat Modelling

Duration 4 weeks

Month Journey

- **Week 1:** Security Architecture and Trust Boundaries **Deliverable**: Asset inventory, trust-boundary diagram and abuse cases.
- **Week 2:** Threat Modelling and Risk Prioritisation. **Deliverable**: STRIDE table, risk register and mitigation-to-test matrix.
- **Week 3:** Identity, Data Protection and Secure Integration. **Deliverable:** Identity flows, access-control matrix and secure integration review.
- **Week 4:** Security Review, Testing and Governance. **Deliverable:** Final security architecture pack and review record.

Final Project Security Architecture and Threat Model Pack

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Month 8

AI Systems Architecture and LLM Application Engineering

Duration 4 weeks

Month Journey

- **Week 1:** AI Application Foundations and Model Integration. **Deliverable**: AI feature brief, model contract, output schema and cost/fallback assumptions.
- **Week 2:** Retrieval-Augmented Generation and Knowledge Grounding. **Deliverable**: RAG diagram, ingestion plan, grounded response contract and evaluation set.
- **Week 3:** Tools, Workflows, Safety and Security. **Deliverable:** Tool contract, workflow state diagram and AI threat-control matrix.
- **Week 4:** Production Readiness, Evaluation and Architecture Review. **Deliverable:** Evaluation report, observability plan and final architecture pack.

Final Project AI Systems Architecture Pack

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Month 9

Advanced AI Agents and Multi-Step Workflow Orchestration

Duration 4 weeks

Month Journey

- **Week 1:** Agent Foundations, Planning and Bounded Autonomy. **Deliverable**: Task graph, state schema and autonomy policy.
- **Week 2:** Tool Orchestration, Permissions and Reliable Execution. **Deliverable**: Tool contracts, permission matrix, idempotency plan and recovery workflow.
- **Week 3:** Memory, Retrieval, Context and Agent Safety. **Deliverable:** Context lifecycle, threat model and human-review design.
- **Week 4:** Agent Evaluation, Observability and Production Readiness. **Deliverable:** Evaluation baseline, trace schema, operational budget and final architecture pack.

Final Project Agent Architecture Pack

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Month 10

AI Engineering Operations, Evaluation and Governance

Duration 4 weeks

Month Journey

- **Week 1:** AI Quality Evaluation and Regression Testing. **Deliverable**: Evaluation rubric, versioned test set, baseline and release thresholds.
- **Week 2:** Versioning, Deployment and Change Control. **Deliverable**: Release manifest, CI gates, staged rollout and rollback plan.
- **Week 3:** Observability, Incidents and Data Governance. **Deliverable:** Monitoring specification, incident analysis, data policy and risk register.
- **Week 4:** Production Readiness and Operational Review. **Deliverable:** Cost/capacity model, runbooks, go/no-go record and final operations pack.

Final Project AI Engineering Operations and Governance Pack

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Month 11

Technical Leadership, Architecture Governance and Engineering Strategy

Duration 4 weeks

Month Journey

Week	Theme	Core output
1	Technical Leadership and Decision-Making	Decision brief and ADR
2	Architecture Governance, Reviews and Technical Debt	Review checklist and debt portfolio
3	Engineering Strategy, Roadmaps and Stakeholder Communication	Roadmap options and stakeholder brief
4	Mentoring, Team Development and Leadership Capstone	Leadership and governance pack, review presentation

Final Project Technical Leadership and Architecture Governance Pack

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Month 12

Engineering Leadership Capstone and Professional Readiness

Duration 4 weeks

Month Journey

Week	Theme	Core output
1	Problem Selection, Scope and Capstone Plan	Project brief, success criteria, scope and plan
2	Architecture Governance, Reviews and Technical Debt	Working increment, architecture and test evidence
3	Hardening, Evaluation and Professional Documentation	Quality review, evidence pack and release candidate
4	Demonstration, Viva and Professional Development	Final release, presentation, viva and development plan

Final Project Three-Year Engineering Capstone